

HAT-P-32 b

R-0002

Practice run · workflow W8-validation · record 2026-07-07_HAT-P-32b_nea · created 2026-07-08T01:22:14+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458107.71353 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1574 +/- 0.0034
Transit depth (Rp/Rs)^2	2.48 +/- 0.11 [%]
Orbital Inclination (inc)	87.88 +/- 0.98
Airmass coefficient 1 (a1)	1.165 +/- 0.0067
Airmass coefficient 2 (a2)	-0.1201 +/- 0.0024
Scatter in the residuals of the lightcurve fit is	0.57 %
Best Comparison Star	None
Optimal Aperture	4.44
Optimal Annulus	7.15
Transit Duration (day)	0.13 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1574 ± 0.0034 vs 0.1489 (deviation 2.29σ)
Duration fit vs published	0.13 d vs 0.1304 d (deviation 0.27σ)
Mid-time O–C	3.6 min
Depth SNR	42.5
Residual scatter	0.57 %

Light curve

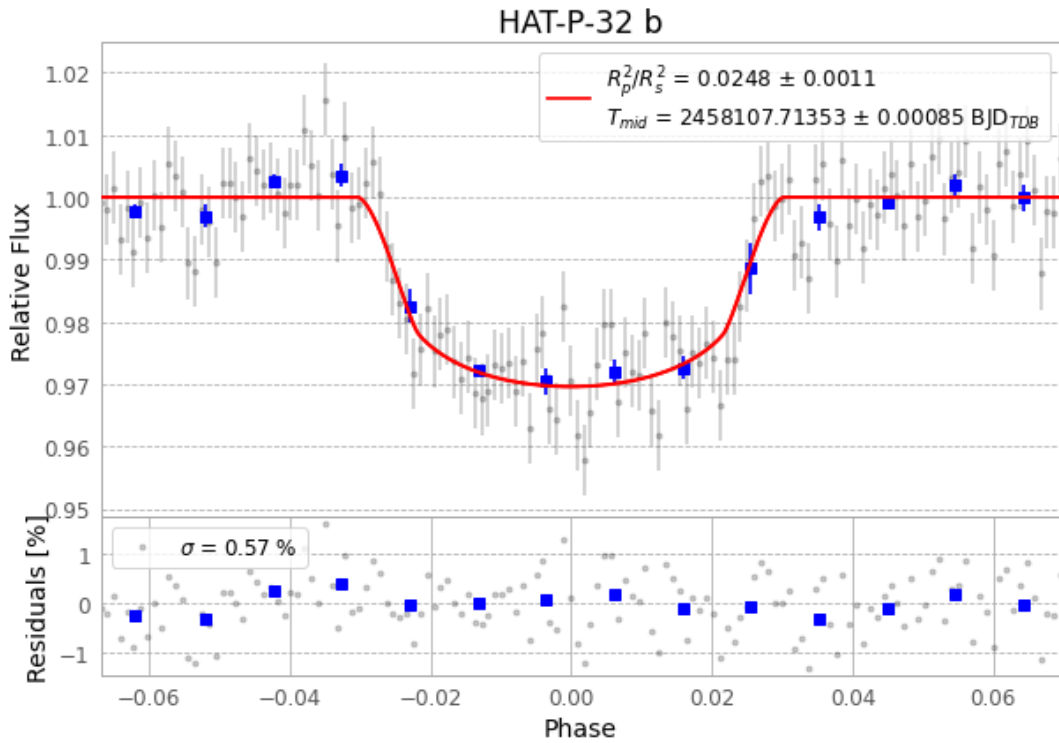


Figure 1 — FinalLightCurve_HAT-P-32 b_17-December-2017.png (SHA-256 9f3e9f2c3d58476b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [424, 286], 56 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 59af1646c0262906...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

142 FITS frames (93 MB), HATP-32171220013343.FITS → HATP-32171220083612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: exotic-nea-test.log (e4c22db1fa96...), FinalLightCurve_HAT-P-32 b_17-December-2017.png (9f3e9f2c3d58...), FinalLightCurve_HAT-P-32 b_17-December-2017.csv (ddd60a1ead48...)